Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 12 GB of memory. The learning rate was set to 0.001 with the Adam optimizer.

2 Results

Figure 1 shows the accuracy during training.

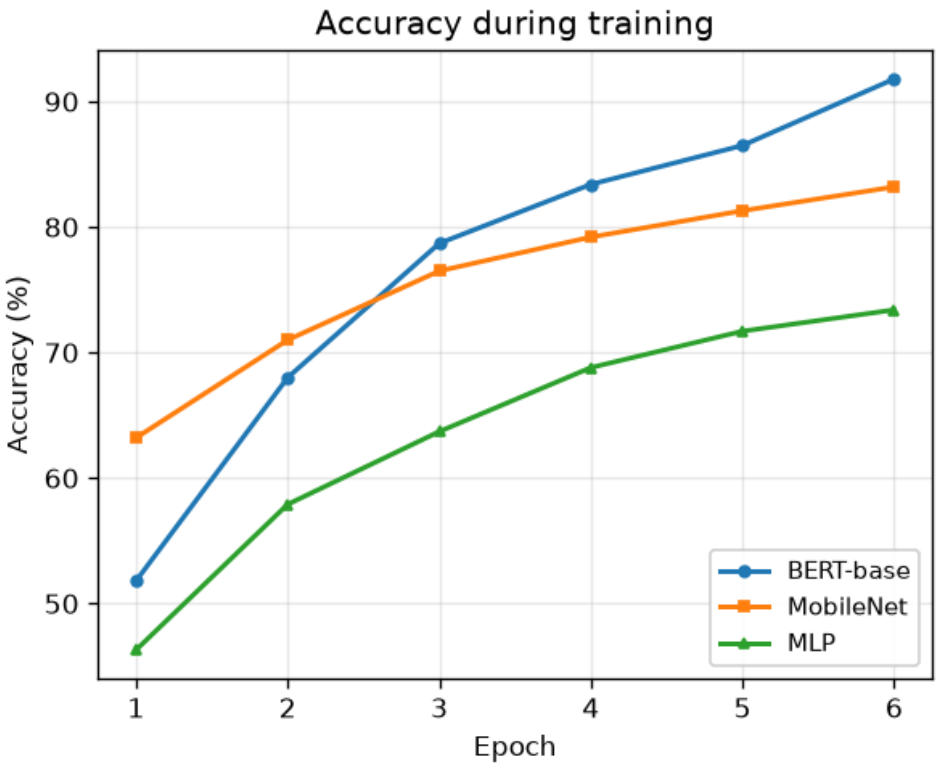


Figure 1: Accuracy during training for the models in this experiment.

At epoch 5, MobileNet achieves 81.3% accuracy. MLP reaches over 55% accuracy at epoch 2. As shown in Figure 1, MobileNet reaches over 80% accuracy at epoch 5. At epoch 4, BERT-base is 6.7 points better than MobileNet.

Figure 2 shows the f1 score during training.

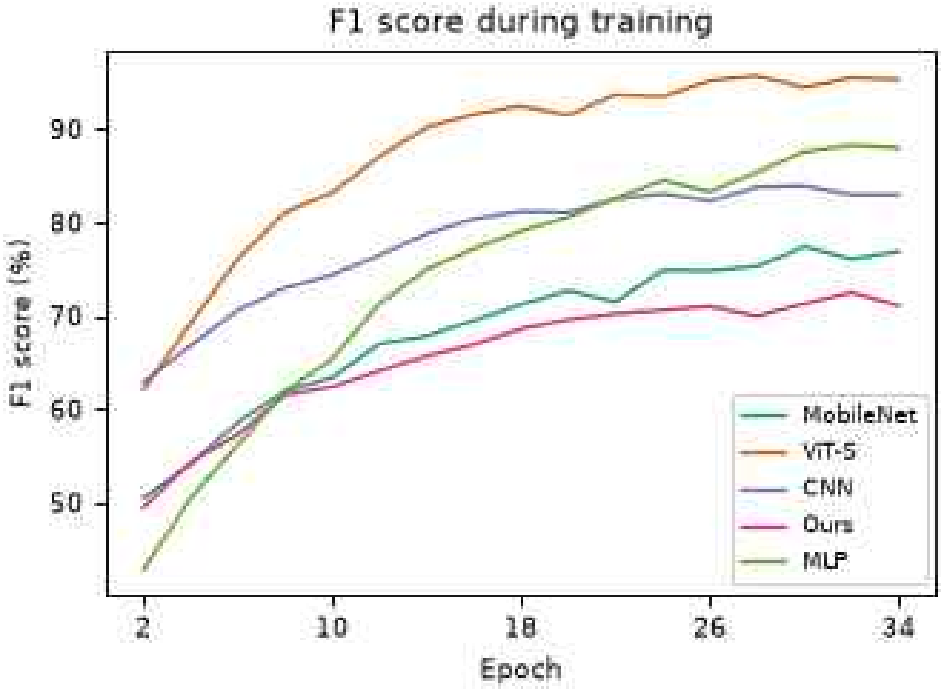


Figure 2: F1 score during training for the models in this experiment.

As shown in Figure 2, Ours reaches about 70% F1 score at epoch 34. At epoch 24, ViT-S achieves 93.5% F1 score. At epoch 28, MobileNet achieves 70.0% F1 score.

Figure 3 shows the training loss.

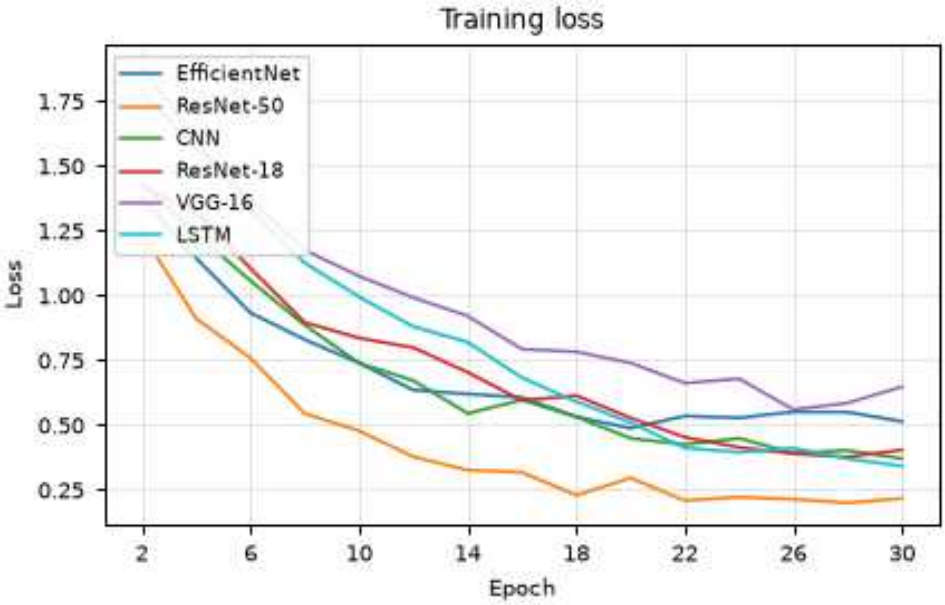


Figure 3: Training loss for the models in this experiment.

VGG-16 has a loss of 1.545 at epoch 8. At epoch 14, the loss of CNN is 1.470 (Figure 3). CNN has a loss of 0.371 at epoch 30. At epoch 14, the loss of EfficientNet is 0.990.

Figure 4 shows the test accuracy during training.

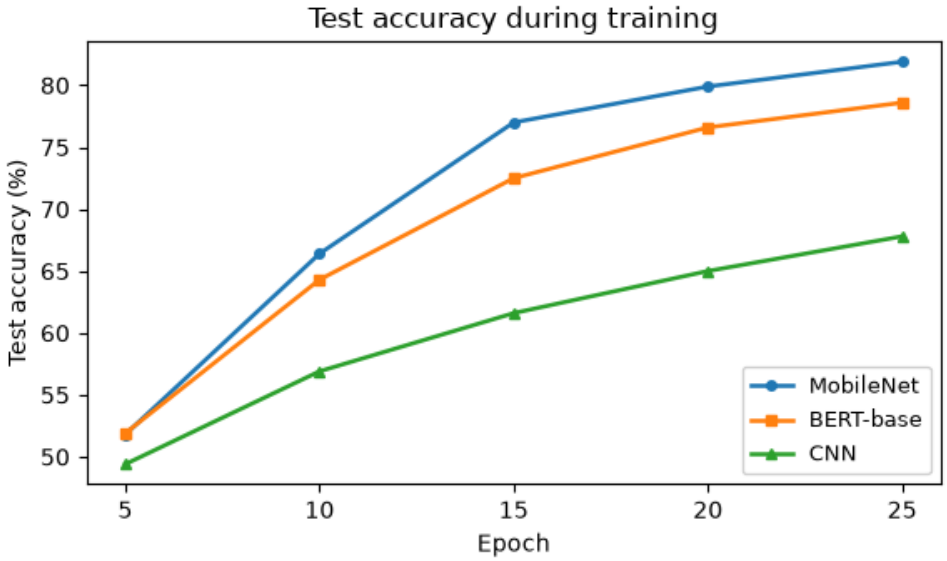


Figure 4: Test accuracy during training for the models in this experiment.

As shown in Figure 4, from epoch 10 to epoch 20, the test accuracy of BERT-base improves by a relative 19.1%. As shown in Figure 4, at epoch 20, MobileNet is 3.3 points better than BERT-base. From epoch 15 to epoch 25, the test accuracy of CNN improves by a relative 11.2%.

3 Discussion

Random Forest reaches 70.2% accuracy on SVHN. We note that the training was stable for all models.